
Image Analysis

An Open-Source Summary

Thomas Debelle



April 20, 2026

Contents

1 Introduction 2

1.1 The basics 2

1.1.1 Light 2

1.1.2 Perception 2

1.1.3 Interaction with matter 2

2 License 6

2.1 Modification 6

2.2 Take down 6

1 Introduction

The key take away is that vision is subjective to the human being and themselves. It is the results of continuous evolution and survival of the fittest. We are more sensitive to light and certain colors, to motion, ... And this fact will be used for efficient image representation and analysis.

We can think of the classic illusion where our brain will interpret certain lengths of an arrow longer than others. Context also does matter, even in a blurry picture our brain will train to make up for what he sees.

1.1 The basics

We first need light to have vision. Light is influenced by object and create perception. ADD image slide 51. The light source bounce on the object and some scattered ray will get trough the 2D Image plane, which will be casted on the sensor array thanks to the lens system.

1.1.1 Light

We will look at it from a physical optic (not geometrical (constant speed and straight path) nor quantum). Light is an Electro-magnetic wave which we assume it to be planar. aka $\frac{|E|}{|H|} = \eta$ in simple term, light is perpendicular. This is a **self-sustaining process** that is characterized by:

1. λ : wavelength
2. Direction
3. E : Amplitude
4. φ : phase
5. Direction of polarisation

Violet: low frequency ($380nm$); Red: high frequency ($760nm$). Remember $\lambda = \frac{c}{f}$.

1.1.2 Perception

We don't perceive light equally. Humans have 3 cones that are not evenly spread apart. But the human vision is trained to compensate for it so everything feels similar in intensity to us. Some animals can have more cones.

1.1.3 Interaction with matter

Table 1: The four types of interaction

Phenomenon	Example
Absorption	Blue water, green leaf
Scattering	blue sky, red sunset
Reflection	colored in k
Refraction	dispersion by a prism

1.1.3.1 Absorption Dissipation of λ specific for the medium. Resonance in molecules.

1.1.3.2 Scattering When light (radiation/particles) deviates from a straight trajectory by local non-uniformities. Type depends on the relative size of particles and wavelengths:

1. **Rayleigh:** small (λ dependent)
2. **Mie:** comparable (weakly λ dependent)
3. **Non-selective:** large (λ independent)

We ignored transmission and diffraction.

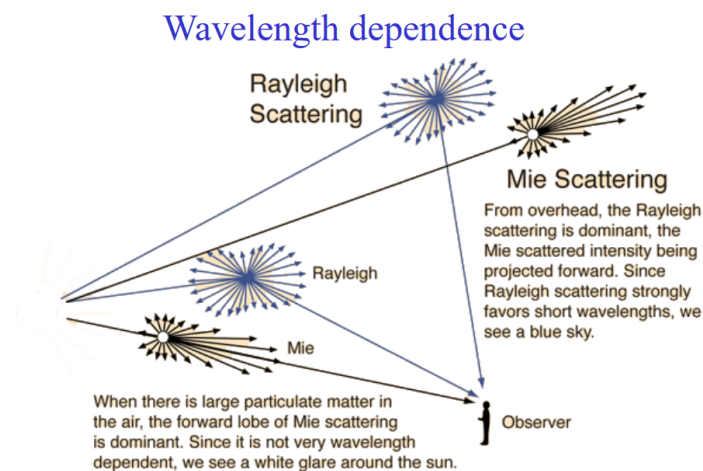


Figure 1: Wavelength dependence for scattering

We need scatter to get the bluesky (Rayleigh with Tyndall effect) or it would be dark. Coloured cloud from volcanic eruption = Mie (particle all over the air). Grey cloud is due to non-selective. Less scatter for lower frequency light (infrared).

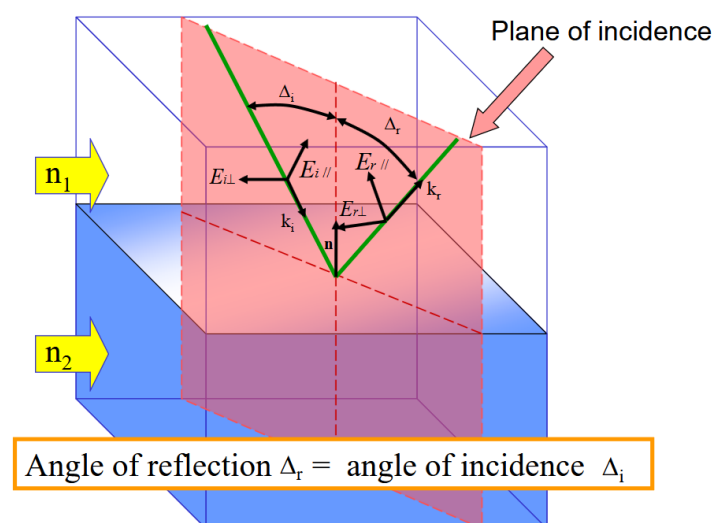


Figure 2: Mirror reflection

1.1.3.3 Reflection With reflection, the notion of polarization is important. The polarization effect creates the fact that a reflected wave on certain surfaces can be reflected with all but one direction of the electric field. This effect is more

prominent on **non-metallic** surfaces!

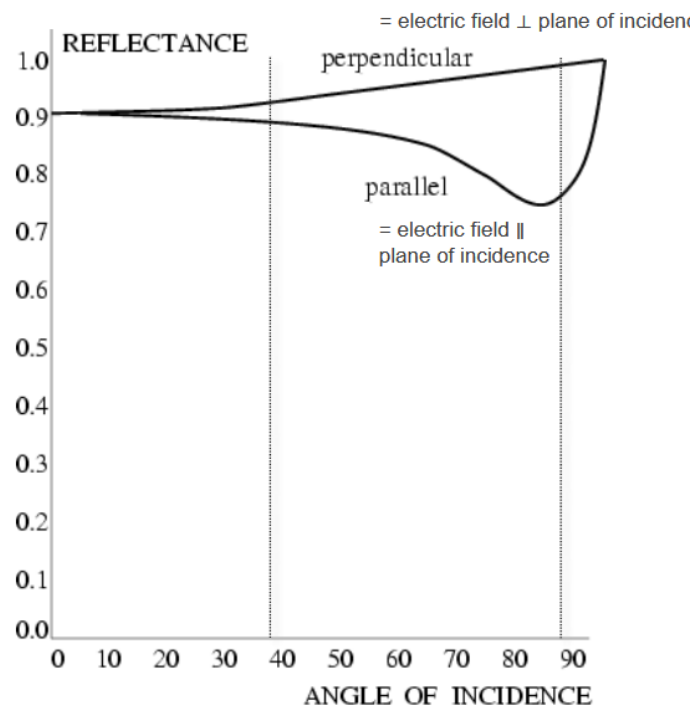


Figure 3: Non-metallic reflection - strong reflectors

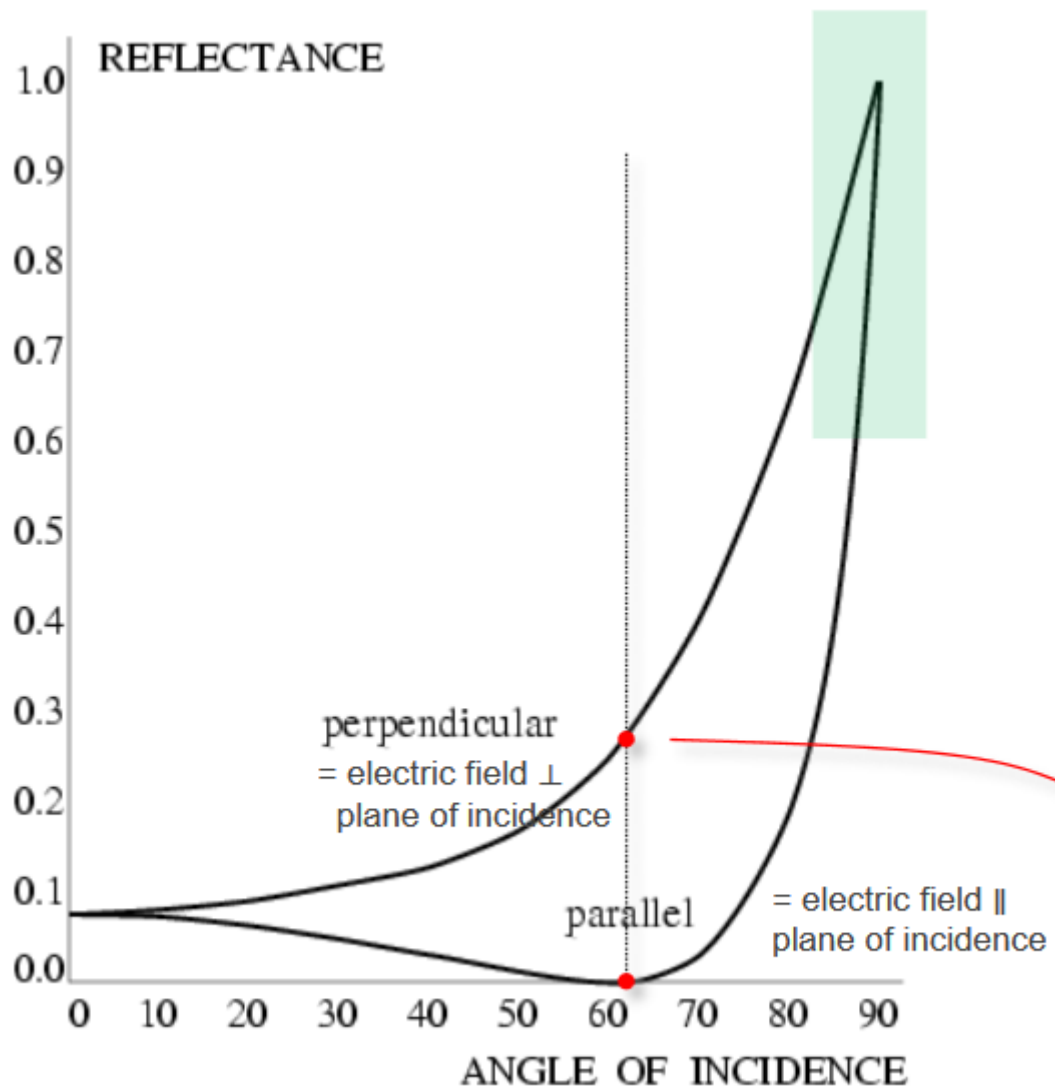


Figure 4: Metallic reflection - Red dot: Brewster angle (polarizer) - Blue zone: Grazing angles

Rough surfaces will lead to “diffuse” reflection as the reflection angles will be all over the place.

2 License

This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0) for KU Leuven students.

You are free to:

- **Share** — copy and redistribute the material in any medium or format with people inside of KU Leuven or being granted access to the source material.
- **Adapt** — remix, transform, and build upon the material if you are a KU Leuven student.

Under the following terms:

- **Attribution** — You must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
- **NonCommercial** — You may not use the material for commercial purposes.
- **ShareAlike** — If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original.

For the full legal text of the license, please visit: <https://creativecommons.org/licenses/by-nc-sa/4.0/legalcode>

2.1 Modification

To contribute to this work or any other one from this project please find more information at the Github repository.

2.2 Take down

If you are a professor or representing any official party and wish to take down this work, you can submit a takedown request at this url: https://github.com/Tfloow/ESATSummary/issues/new?template=takedown_notice.yml

Some Rights Reserved 2026 Authors of the Summary, Professors of the Course and possible book's authors. Some Rights Reserved.